Sham Satish Thakare

Candidate for Master of Engineering (ME) · Distributed Systems Architecture, Modern C++20 & Cloud Telemetry

Pre-Doctoral Researcher, Capital One Systems Research (Mentored by Yuri Shelke, Distinguished Engineer) · APCOER Pune (SPPU)

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SYSTEMS & ENGINEERING OBJECTIVES

Distributed Systems Architecture: High-throughput state-machine replication (Raft, Paxos), dynamic quorum adaptation under network partitioning, modern C++20 lock-free concurrent programming, and tail latency minimization.
Cloud Platforms & Telemetry: Multi-region infrastructure, failure-domain aware routing, automated traffic management, and high-assurance runtime interception systems.
Engineering Leadership: University leadership as President of the Social Volunteer Club (leading 250+ volunteers) and certified architectural credentials.

EDUCATION & HONORS

APCOER Pune (Savitribai Phule Pune University), Pune, India — 2020 – 2024

Bachelor of Engineering (B.E.) in AI & DS | CGPA: 8.70 / 10.00 (First Class with Distinction) — Final Year SGPA: 9.53 / 10.00 (Rank 1)

Gold Medal in Mathematics (2024): Conferred by university leadership for academic excellence across quantitative coursework.

Grade O (Outstanding 10/10) Systems Coursework: Distributed Computing (Course 417531), Operating Systems (Course 217521), Cyber Security (Course 317530), Senior Capstone Project, Industry Internship.

Medium of Instruction: Verified on official university transcripts as **ENGLISH** (Ref: EXAM./TRANS./BEN241203044).

Pre-University: Class XII HSC (71.23%, Seat: V001345) • Class X SSC (**85.20% Distinction, Mathematics: 95/100**, Seat: H006786).

PROFESSIONAL CERTIFICATIONS & TECHNICAL EXPERTISE

Oracle Cloud Infrastructure 2024 Certified Architect Associate (Cert ID: 100788330OCI24CAA, Validation: b962e71fa08d4b31a39ae78eb486bfef).

Oracle Cloud Infrastructure 2024 Certified Generative AI Professional (Cert ID: 100788330OCI24GAIPC, Validation: c4e357908b9848e0bf24775d1f86d6ee).

Microsoft Certified: Azure AI Fundamentals & SQL AI Developer (Verifiable Credential via Microsoft Learn).

Google Certified: AI Professional (Credential verified).

IIT Bombay Spoken Tutorial: Kotlin Honors (Exam Certificate verified).

Languages & Frameworks: Modern C++ (C++20, coroutines, atomics, concepts), Python 3.12, Go, SQL, Linux Kernel / eBPF, Git, Docker, Kubernetes, AWS, OCI.

SYSTEMS RESEARCH & SOFTWARE ARCHITECTURE

Capital One Systems Research, Pune, India — 06/2024 – Present

Pre-Doctoral Researcher | Mentorship: Yuri Shelke (Distinguished Engineer)

Architected and implemented <code>quorumshift</code> in modern C++20: dynamic quorum re-weighting Raft consensus engine that eliminates p99 tail latency spikes during asymmetric cloud network partitions.

Designed asynchronous, lock-free distributed benchmarking harness evaluating multi-node consensus throughput across heterogeneous network partitions and emulated cross-datacenter WAN topologies.

Formulated multi-stage deterministic runtime policy interception and out-of-band schema validation barriers in <code>medirush-safeagent</code>, guaranteeing zero unhandled side-effects.

APCOER Pune (Savitribai Phule Pune University), Pune, India — 2023 – 2024

President, Social Volunteer Club (Institutional Leadership)

Elected President of university student volunteer organization; mobilized and led 250+ student volunteers across regional educational outreach, health camps, and social welfare programs impacting 1,200+ beneficiaries across Maharashtra.

TARGET HARVARD SEAS FACULTY ALIGNMENTS

Prof. Minlan Yu (SEAS): High-speed networked systems, cloud telemetry, automated traffic management, and datacenter infrastructure.

Prof. James Mickens (SEAS): Fault-tolerant distributed systems, state-machine replication, and resilient computing.

Prof. Stratos Idreos (SEAS): Scalable data systems architecture, memory-efficient data structures, and DASL laboratory.